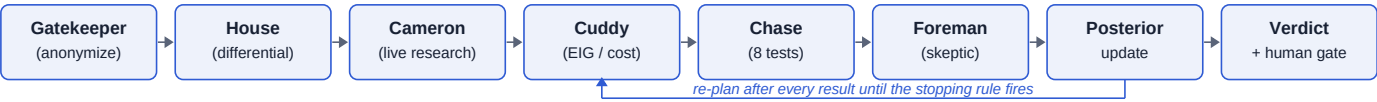


Princeton-Plainsboro: an agentic lab for exoplanet candidate diagnosis

Hack-Nation Global AI Hackathon · Challenge 03: Agentic Scientific Discovery · Team HousePhD



Challenge tackled

- Telescopes flag thousands of transit-like dips; most are eclipsing binaries, starspots, blends or glitches. The bottleneck is choosing **which vetting test to run next and when to stop**.
- **User:** exoplanet survey scientists and vetting teams who face more candidates than people.

Tools / ML models used

- **Gemini (flash) or Claude Sonnet:** Cameron reads live papers and writes cited briefs; optional agent voices.
- **arXiv + OpenAlex APIs:** live literature search; every citation is resolved before use.
- **Omnigent:** agent bundle (House root + 5 sub-agents), tools, guardrail policies.
- **NumPy / SciPy:** 8 light-curve tests; Bayesian posterior; expected information gain (EIG).
- **Streamlit:** live demo UI with human approval gates. **NASA Exoplanet Archive + MAST (lightkurve):** 48 real Kepler candidates.

How we spent our time (in order)

1. Problem framing, hypothesis space H1–H5, agent roles and policy design
2. Synthetic light-curve generator and the 8 vetting tests
3. Likelihood calibration, EIG planner, agent loop, whiteboard and ledger
4. Benchmark (baselines, ablations, oracle), real-Kepler transfer test
5. Omnigent bundle + live run, Streamlit UI, live LLM research, submission

What worked well

- **1.9× lower test cost** than a fixed checklist at matched accuracy (240 blind synthetic targets, 90% CI 1.47–2.39).
- Accuracy 0.917 vs 0.846 (checklist); Brier 0.133 vs 0.243; human escalations 42% → 26%.
- Leak-proof evaluation: labels held only by a Data Gatekeeper, **0 agent label reads** in the audit.
- Live research guardrail: LLM citations outside the retrieved set are stripped automatically.

What was challenging

- **Synthetic → real transfer.** On real Kepler data accuracy fell to 0.38–0.54. Leave-one-out recalibration cut confident errors from 11 to 2; we report this as a negative result.
- **Correlated tests.** Tests share one fit, so posteriors were overconfident; added CV-fitted tempering.
- **LLM reliability.** Invented citations and leaked drafting text; fixed with JSON output plus an ID allow-list.
- **Free-tier quotas.** Gemini models hit daily limits; added automatic fallback across models.

Results

Condition (240 blind targets)	Accuracy @0.9	Mean cost	Brier	To human
Fixed checklist	0.846	6.62	0.243	42%
Planner only (no debate)	0.883	5.56	0.159	24%
Full House team	0.917	6.62	0.133	26%
Real Kepler (48 KOIs, transfer)	0.38–0.54	—	—	96% after recal.

Next: If we had 24 more hours, we'd calibrate on ~2,000 real DR25 Kepler candidates, learn a joint likelihood instead of assuming independent tests, and re-measure the speedup on real and TESS data.

Data & references: NASA Exoplanet Archive KOI cumulative table; MAST Kepler light curves; method papers resolved on arXiv (Coughlin+2016, Thompson+2018, Seager & Mallén-Ornelas 2003, Bryson+2013, McQuillan+2014).